

GAME DESIGN DOCUMENT

CARGO INDUSTRY

Transportation and industrial management strategy

Single-player • PC (Windows) • Unity

DOCUMENT	Game Design Document
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About this revision

Version 0.1 of this document described the game's systems but carried almost no numbers, because none were available. Version 0.2 incorporated a value analysis extracted from the project source, closing most of those gaps: construction costs, vehicle specifications, production and decay rates, the wage formula, movement physics, the accounting cycle and the full production chain.

Version 0.3 adds `ResourcePricing`, and with it the revenue side of the economy: the sale formula, the complete price list for both transport modes, and the analysis that follows from having costs and income in the same document for the first time. Section 8.4 and the revised throughput figures in section 9 are new, as are three findings that only become visible once prices are known.

Two markers remain in use:

`[TBD]` marks a value that could not be determined, either because the class holding it was not among the analysed files or because it is set in the Unity Inspector rather than in code.

(verify) marks a statement derived by inference rather than read directly from source.

One caveat applies throughout. Values marked `[SerializeField]` in Unity can be overridden in the Inspector, and Unity retains previously serialised values when a script is replaced. Every number below is the default found in code. Where a scene value differs, the scene wins.

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1 Design overview

1.1 High concept

Cargo Industry is a single-player transportation and industrial management strategy game. The player builds a railway and road network across a procedurally generated world, decides where industry is located, staffs each facility, and earns revenue by moving raw materials and finished goods between production, processing and destination locations.

1.2 Design pillars

1. **The player authors the industrial map.** Factories are not seeded by the generator. Where each facility stands, and which type it is, is a player decision constrained by terrain rather than by the designer.
2. **The network is the game.** Every mechanic exists to make the transport network either work better or work worse. Nothing sits outside that loop.
3. **Iteration is not punished.** Automatic rerouting and free-form diagonal track exist so that rebuilding a mature network is a normal activity rather than a setback.
4. **Legibility over spectacle.** Complexity is acceptable; hidden state is not.
5. **A different world every time.** Replay value comes from the generated map invalidating the previous optimal layout.

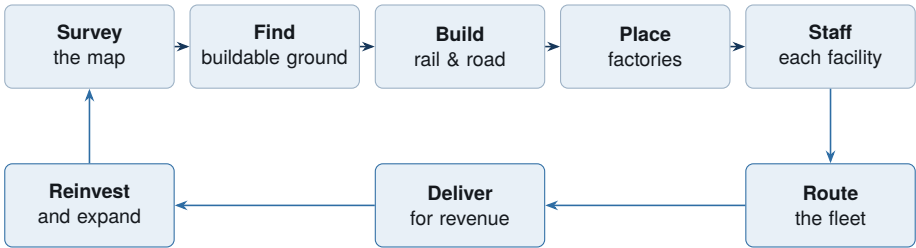
1.3 Non-goals

- **Passenger transport.** Cargo only: raw materials and manufactured goods.
- **Combat and failure states.** No combat, no timer, no loss condition. The balance may go negative without the game ending.
- **Narrative.** No campaign, protagonist or dialogue.
- **Multiplayer.** Single-player.
- **Competing AI companies.** None appear in the analysed systems. (*no AI competitor logic is present in the economy or fleet code*)

1.4 Reference titles

Transport Tycoon (1994, MicroProse) for the transport and infrastructure structure; **Industry Giant** (1997, JoWooD) for the industrial and production concept. The design departs from both in one respect: in those games the industrial map is given and the player serves it. Here the player builds it.

2 Core gameplay loop



2.1 Session shape and time base

The game has no defined end state. A session runs continuously on a calendar beginning 1 January 1950.

TIME BASE	3 real seconds = 1 game day
ONE GAME MONTH	28–31 days = 84–93 real seconds (≈1.5 min)
ONE GAME YEAR	365 days = 1 095 real seconds (≈18.25 min)
LEAP YEARS	Handled correctly via <code>DateTime.DaysInMonth</code>
DISPLAY FORMAT	“January, 1950”
CATCH-UP ON LAG	Missed days are processed individually in a loop

Three clock events drive the simulation: `OnDayElapsed` for daily production and fleet lifecycle, `OnMonthElapsed` for wages, operating costs, accounting and the price multiplier, and `OnDateChanged` for UI redraw.

3 Game world

3.1 Grid and terrain

GRID SIZE	256 × 256 tiles
TILE WORLD SIZE	1.0
BASE HEIGHT	2.75 world units
ELEVATION STEP	0.25 world units
ELEVATION LEVELS	−1 to 10 (world Y 2.75 to 5.50)
PROTECTED BORDER	1 ring of tiles at the map edge
WATER LEVEL	Fixed at Y = 2.75

3.1.1 Generation

Terrain is generated with layered Perlin noise.

TERRAIN SEED	0
NOISE SCALE	0.03
OCTAVES	4
PERSISTENCE	0.5
LACUNARITY	2.0
SEA THRESHOLD	0.30
SEA FLOOR LEVEL	−2
MAX LAND LEVEL	8
FLAT BIAS	1.5

Each tile is a quad built from four corner vertices; heights inside a tile are bilinearly interpolated. Triangulation is chosen according to whether opposite corners share a height, which avoids visible seams on sloped ground.

Two things to reconcile. The sea floor level is −2 while the minimum elevation level is −1; confirm which bound applies in practice. Separately, the terrain overlay presents five elevation bands to the player, whereas the engine works in twelve discrete levels. The legend is a deliberate simplification, but it should be recorded as such rather than read as the underlying model.

3.2 Cities

CITY SEED	0 (different cities each run)
CITIES PER MAP	25 to 30
REGION SIZE	10 to 15 tiles
SPACING BETWEEN CITIES	1 free tile minimum
PLACEMENT ATTEMPTS	80 per city
BUILDING DENSITY	0.30 (30 % of region tiles built on)
TOWN TYPE WEIGHTS	Small 0.35 • Medium 0.40 • Big 0.25
BUILD LEVELS	0 to 5
MAX STATIONS PER CITY	20

3.2.1 Silhouette rules

Building height follows town tier: Small towns use one tile width, Medium two, Big four. Within a Big Town, high buildings occupy the inner 34 % of the region radius and at least medium buildings the inner 67 %; in a Medium Town the inner 55 % carries medium buildings. A jitter of 0.12 softens the band edges. Gaps between buildings never occur at the centre and occur 35 % of the time at the edge. The net effect is a readable density gradient from a dense core to a ragged periphery.

The building library provides three tiers of five variants. Prefabs are optional; where none is assigned a coloured primitive box is generated.

Cities remain economically undefined. The generator is fully documented, but nothing in the analysed systems shows cities consuming goods, generating demand or growing. On present evidence a city is a named delivery location and a source of station names rather than an economic actor. This should be confirmed, because it is the first question a genre-literate player or reviewer will ask. See §14.



Generated terrain: water at fixed $Y = 2.75$, woodland and rock. Trees and rocks block construction and can be demolished free of charge.

4 Construction and costs

4.1 Build costs

Rail and road use **identical price lists**. Every cost below is deducted through `TrySpendCredits`, which fails if the player cannot cover it, so construction is always fully funded.

ELEMENT	COST	REFUND
Terrain level up / level down	10 CR	5 CR
Straight piece (horizontal / vertical)	25 CR	12 CR
Curve (4 variants)	25 CR	12 CR
Switch (4 variants)	25 CR	12 CR
Crossroad	25 CR	12 CR
Station (horizontal / vertical)	30 CR	15 CR
Depot (4 orientations)	50 CR	25 CR
Extraction facility (any of 7)	500 CR	250 CR
Processing facility (any of 9)	300 CR	150 CR
Trees, rocks, landing locations	not buildable	0 CR

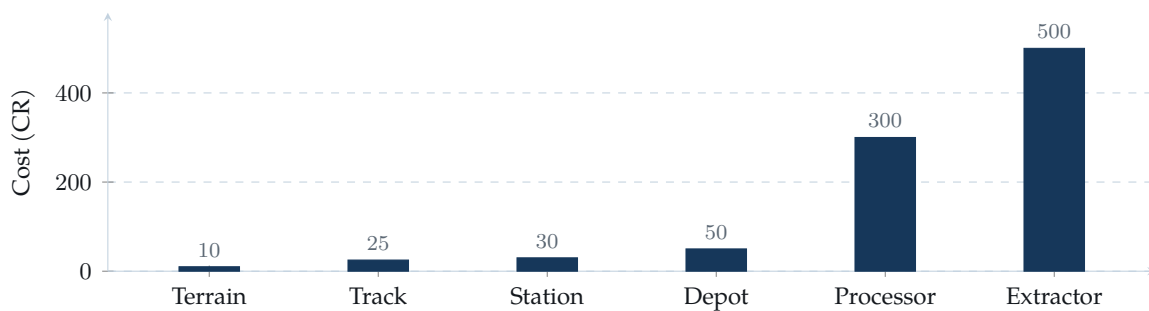


Figure 1 — Build costs. Infrastructure is close to free relative to facilities; an extraction site costs the equivalent of twenty track pieces.

4.2 Demolition refund

The rule is universal: **50 % of the base build cost, rounded down** by integer division. A demolished track piece returns 12 of the 25 spent, so the player loses one credit to rounding on every piece. A facility is refunded *once for the whole object*, not per footprint tile. Empty tiles and environment objects refund nothing, and landing locations cannot be demolished at all.

4.3 Footprints and exclusion zones

FACILITY	FOOTPRINT	FACILITY	FOOTPRINT
Coal Mine	2×3	Power Station	2×2
Forest	3×3	Sawmill	2×2
Iron Ore Mine	3×3	Oil Refinery	2×2
Gold Mine	3×3	Electronics Factory	3×3
Silver Mine	3×3	Furniture Factory	3×3
Farm	3×3	Slaughterhouse	2×2
Oil Wells	3×3	Grain Factory	2×3
		Smelter	2×3
		Glass Factory	2×2

Facilities rotate in 90° steps; 90° and 270° swap width and depth. Unknown types default to 1×1.

4.3.1 Exclusion radii

Both are Chebyshev distances.

- **Station exclusion radius 2** — a 5×5 region around a station, so two stations need at least two free tiles between them.
- **Factory exclusion radius 3** — measured *from the edge of the footprint*, so two facilities need at least three free tiles between them.

Combined with the terrain blockers, this is what makes siting a search rather than a formality: a 3×3 facility with a 3-tile margin effectively reserves a 9×9 area of suitable ground.

4.4 Construction time

Facility construction runs on a real-time timer driven by `Time.deltaTime`, with progress shown as Building process: NN%. The label lingers 2 seconds after completion. During construction the facility can be neither edited nor demolished.

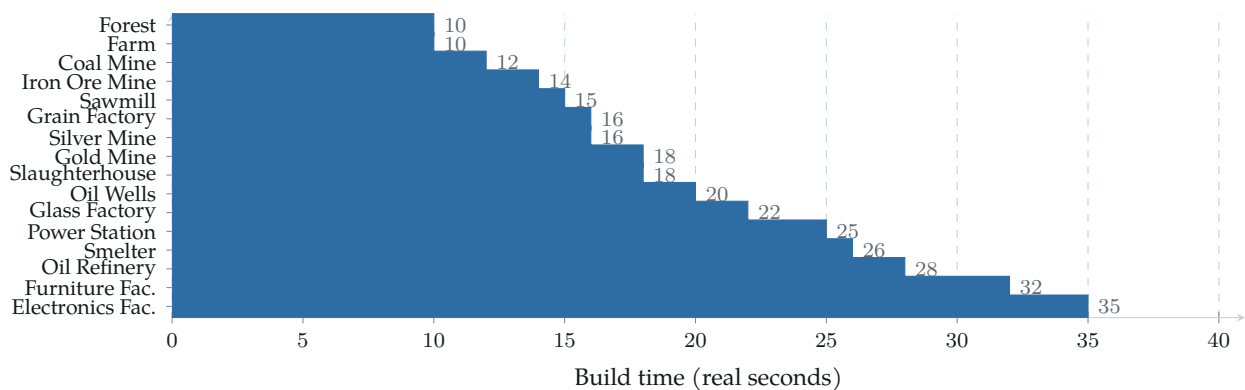


Figure 2 — Facility construction times. Deeper positions in the production chain take longer to build, which reads as intentional pacing.

Note the unit mismatch. Construction time is the only major system measured in real seconds; production, wages, servicing and accounting all run on the game clock. At 3 real seconds per game day, a 35-second Electronics Factory build spans roughly 12 game days — but that relationship breaks if the time base is ever changed.

5 Industry and production

5.1 Facility catalogue

Sixteen facility types exist in two sets. Set 1 (tile ID 4) extracts raw materials and has no inputs. Set 2 (tile ID 5) processes inputs into goods.

FACILITY	ID	PROFESSION	INPUTS (CAPACITY)	OUTPUTS (CAPACITY)
Set 1 — extraction, 500 CR, no inputs				
Coal Mine	0	Miner	—	Coal 10 000
Forest	1	Woodcutter	—	Wood 10 000
Iron Ore Mine	2	Iron Miner	—	Iron Ore 10 000
Gold Mine	3	Gold Miner	—	Gold 10 000
Silver Mine	4	Silver Miner	—	Silver 10 000
Farm	5	Farmer	—	Livestock 9 000; Grain 12 000
Oil Wells	6	Oil Driller	—	Oil 15 000
Set 2 — processing, 300 CR				
Power Station	7	Power Plant Opr.	Coal 17 000	— (terminal sink)
Sawmill	8	Sawyer	Wood 7 500	Boards 8 500
Oil Refinery	9	Refinery Operator	Oil 12 000	Plastic 7 500
Electronics factory	10	Electronics Tech	Plastic 9 500; Metals 9 500	Glass 12 500 Electronics 12 500
Furniture Factory	11	Carpenter	Plastic 12 000; Metals 12 000; Boards 12 000	Glass 12 000; Furniture 18 000
Slaughterhouse	12	Butcher	Livestock 7 750	Meat 11 000
Grain Factory	13	Miller	Grain 7 500	Flour 9 000
Smelter	14	Metallurgist	Iron Ore 9 000	Metals 12 000
Glass Factory	15	Glassmaker	Metals 13 000	Glass 12 000

(The facility-to-profession mapping follows from the two in-game lists, which contain exactly matching counts in matching categories.)

5.2 Resource register

Resource identifiers are shared with wagon and vehicle identifiers, so the type of a transport unit determines its cargo with no additional field.

ID	RESOURCE	ID	RESOURCE	ID	RESOURCE
0	None	6	Livestock	12	Flour
1	Coal	7	Grain	13	Metals
2	Wood	8	Oil	14	Glass
3	Iron Ore	9	Boards	15	Furniture
4	Gold	10	Plastic	16	Electronics Products
5	Silver	11	Meat		

5.3 Production chain

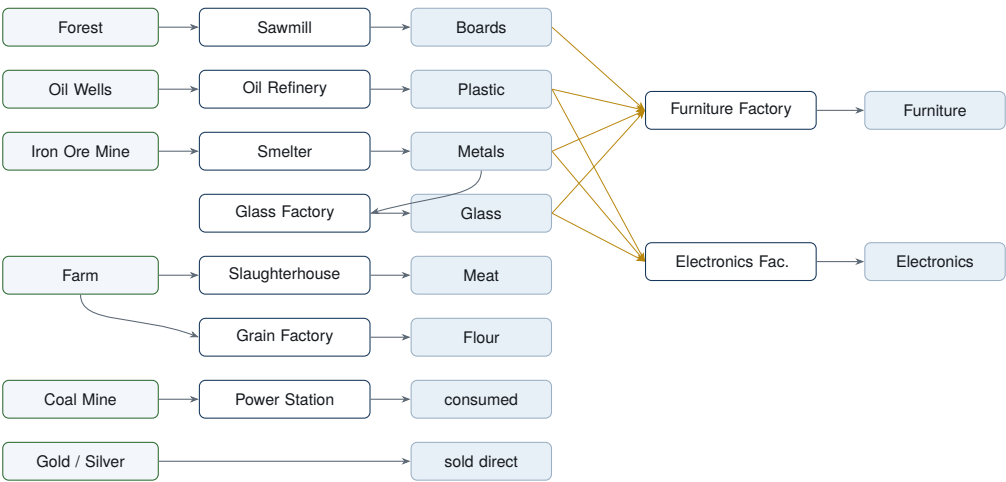


Figure 3 — The full production graph. Green borders are extraction, navy are processing. Amber arrows are the tier-two feeds: Furniture consumes Boards, Plastic, Glass and Metals; Electronics consumes Plastic, Glass and Metals.

5.3.1 Structural observations

- **Metals is the busiest intermediate.** It feeds the Glass Factory, the Electronics Factory and the Furniture Factory, so the Iron Ore Mine to Smelter link is the most load-bearing part of any mature network.
- **Glass is made from Metals, not sand.** An unusual chain worth a deliberate decision: accept it as a stylised abstraction, or revisit it, because players familiar with the genre will notice.
- **Gold and Silver have no processor.** No facility in the game lists either as an input. Together with Meat, Flour, Furniture and Electronics, they form a group of six resources with no consumer — see the warning in §8.4.

- **The Power Station is a pure sink.** It consumes Coal and produces nothing, giving Coal an endpoint without a further chain.
- **The Farm is the only dual-output producer,** yielding both Livestock and Grain and therefore feeding two processors from one site.
- **Chain depth is three at most:** Iron Ore → Metals → Glass → Furniture or Electronics.

5.4 Production rate

Production runs once per game day and is governed entirely by the facility's salary level.

$$\text{amount} = \min(\text{capacity}, \text{amount} + \text{ProductionPerDay}(\text{LevelSalary}))$$

All three conditions must hold: the facility is in Set 1, its occupancy flag is true, and construction is complete.

LEVEL	UNITS/DAY	UNITS/REAL MIN	PROD. INDEX	WAGE INDEX
0	3	60	1.00×	1.0×
1	6	120	2.00×	1.5×
2	10	200	3.33×	2.0×

Set 2 facilities do not produce on a timer; their output slots fill only as a consequence of deliveries.

5.5 Decay

If a processing facility's slot sits at full capacity, unchanged, for **three consecutive game months**, decay begins at **−10 units per game day** until the slot empties and can receive again.

Decay does not count as a change for wage purposes, so it cannot by itself trigger a salary payment. Set 1 facilities do not decay; at full capacity they simply stop producing.



The factory catalogue in game, split into producers and processors.

6 Staff and wages

6.1 Core rule

A facility with no assigned worker does not operate. Occupancy is a precondition of production, and of any wage being paid.

6.2 Wage formula

$$\text{Monthly wage} = \text{EmployeeSalary} \times k(\text{LevelSalary}), \quad k = \{1.0, 1.5, 2.0\}$$

The level-1 result is rounded. If $\text{EmployeeSalary} \leq 0$ the wage is zero. The wage is charged once per facility regardless of slot count, and is never prorated.

6.2.1 Payment condition

The full wage is deducted only if, during the elapsed game month, *both* held: the facility was occupied at least once, *and* at least one slot amount changed, whether through production or through a trade. Otherwise the wage is zero, so an idle facility is free to keep.

6.3 The dual effect of LevelSalary

A single parameter raises both production and wage, but not at the same rate.

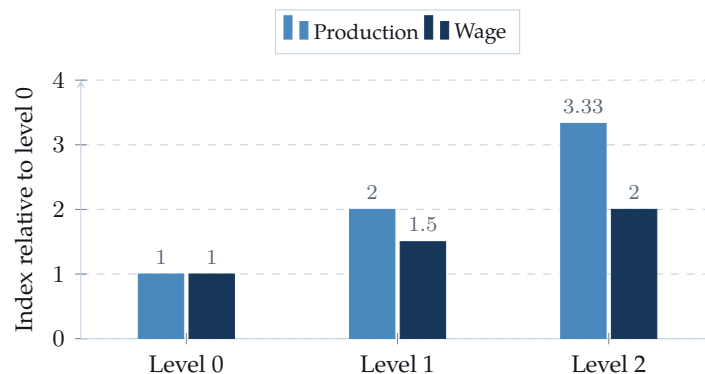


Figure 4 — Production rises faster than wage at every step. Units produced per unit of wage: 3.00 at level 0, 4.00 at level 1, 5.00 at level 2.

Balance finding: the salary level is not a trade-off. Production scales 1 : 2 : 3.33 while wage scales 1 : 1.5 : 2, so level 2 returns 67% more output per credit of wage than level 0. No configuration exists in which a lower level is preferable, provided the output can be hauled away. The selector therefore presents a choice with only one correct answer.

This is worth addressing, because the staffing system is otherwise a genuine differentiator for the game. Options include a non-linear cost at higher levels, a hiring or availability constraint on skilled staff, or tying the level to a risk the player must manage.

Possible exploit: production may be free at any level. Production depends on LevelSalary and the occupancy flag. The wage depends on EmployeeSalary and is zero when that value is zero or below — which is also its default on placement. As documented, a facility set to LevelSalary = 2 with EmployeeSalary = 0 would produce at the maximum rate of 10 units per day at no cost.

StaffManagement exposes both fields as public and freely mutable, with no construction guard and no read-only accessor, unlike FactoryInstance where the ID is fixed. *(Confirm whether the interface constrains the value, for example by enforcing a minimum salary per level. If it does not, this is the single most significant balance hole in the economy.)*



The staff screen: profession roster, portrait, role description and the three salary tiers.

7 Transport and fleet

7.1 Locomotives

	COST	OP ./MO	SPEED	WEIGHT	POWER	LIFE (Y)	SERVICE (D)
NAME							
Iron Dragon	100	5	60	150	60	8	1 500
Desert Runner	150	7	100	200	100	15	2 000
Thunderbolt	200	10	150	200	150	20	2 500

Year of manufacture is recorded as 1930, 1970 and 2000 respectively. *(Whether this gates availability by date is not established in the analysed files; see §14.)*

7.2 Wagons

All sixteen wagon types cost **10 CR** and carry **15 units**. They differ only in weight: Coal Truck 5, Wood Truck 7, and 2 for the remaining fourteen.

$$\text{Train cost} = \text{locomotive} + \text{wagonCount} \times 10 \text{ CR}, \quad \text{wagonCount} \in [1, 10], \text{ default } 5$$

A five-wagon Iron Dragon costs 150 CR and carries 75 units; a ten-wagon Thunderbolt costs 300 CR and carries 150.

7.3 Road vehicles

All sixteen share cost 50 CR, operating cost 3 CR per month, capacity 20 units, weight 2 and a service life of 10 years. They differ in speed and service interval.

VEHICLE	SPEED	SERVICE (D)	VEHICLE	SPEED	SERVICE (D)
Coal Truck	60	2 150	Boards Truck	80	3 350
Wood Truck	60	2 300	Plastic Truck	80	3 500
Iron Ore Truck	60	2 450	Meat Truck	80	3 650
Gold Truck	60	2 600	Flour Truck	80	3 800
Silver Truck	60	2 750	Metals Truck	60	3 950
Livestock Truck	45	2 900	Glass Truck	45	4 100
Grain Truck	60	3 050	Furniture Truck	80	4 250
Oil Truck	45	3 200	Electronics Truck	80	4 400

The service interval rises linearly by 150 days per vehicle ID, from 2 150 to 4 400. A road vehicle is a single body with no wagons and carries its capacity itself.

7.4 Movement physics

Constants are identical for trains and road vehicles.

MOVE TIME	2.0 s per tile at reference speed
SPEED REFERENCE	100
STATION WAIT	10.0 s
TRADE DELAY	2.0 s after arrival
BREAK WAIT	1.0 s when the route is interrupted
ACCELERATION TUNING	0.02
BRAKE MULTIPLIER	1.6
MINIMUM ACCELERATION	0.15
UPHILL SENSITIVITY	1.3, minimum 35 % of max speed
DOWNHILL SENSITIVITY	0.6, maximum 135 % of max speed
PATHFINDING	A*, iteration cap 10 000, 0.4 s debounce after map change
CONSIST SPACING	0.85 between members

$$v(\text{Speed}) = \frac{\text{Speed}}{100} \times \frac{1}{2.0} \text{ tiles/s}$$

$$a = \max\left(0.15, \frac{\text{Power}}{\text{total weight}} \times 0.02\right)$$

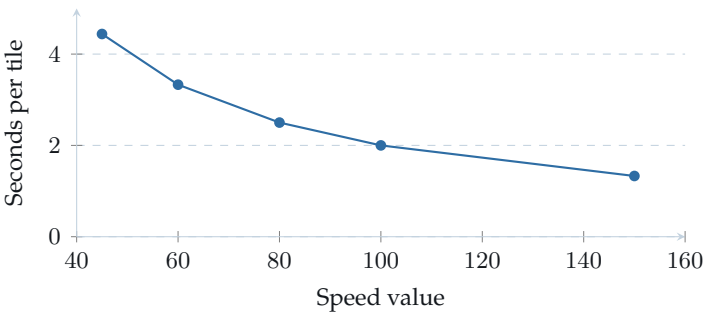


Figure 5 — Traversal time per tile. The relationship is inverse, so gains flatten sharply above Speed 100.

Balance finding: the acceleration model never engages. The floor of 0.15 is exceeded only when $\text{Power} / \text{weight} > 7.5$. The most favourable ratio in the game is a Thunderbolt with no wagons at $150/200 = 0.75$ — an order of magnitude short. Every train and every road vehicle therefore accelerates at exactly 0.15 and brakes at exactly 0.24, in all configurations. The practical consequence is that **wagon weight has no effect on anything**. The distinction between the Coal Truck at weight 5, the Wood Truck at 7 and the other fourteen at 2 is inert, as is locomotive power as a differentiator. If heavy cargo is meant to feel heavy, either ACCEL_TUNING must rise by roughly an order of magnitude, or the floor must fall.

7.5 Depots, servicing and lifetime

One active vehicle per depot. A vehicle's identity is its depot coordinates, so fleet size is bounded by depot count and depot placement is a network-planning decision.

Servicing and lifetime are evaluated every game day. The service interval, in game days, recurs for the whole life of the vehicle: on reaching it the vehicle returns to its depot and the counter resets. The service life, in whole calendar years since creation, fires once, sends the vehicle to the depot and is then marked handled.

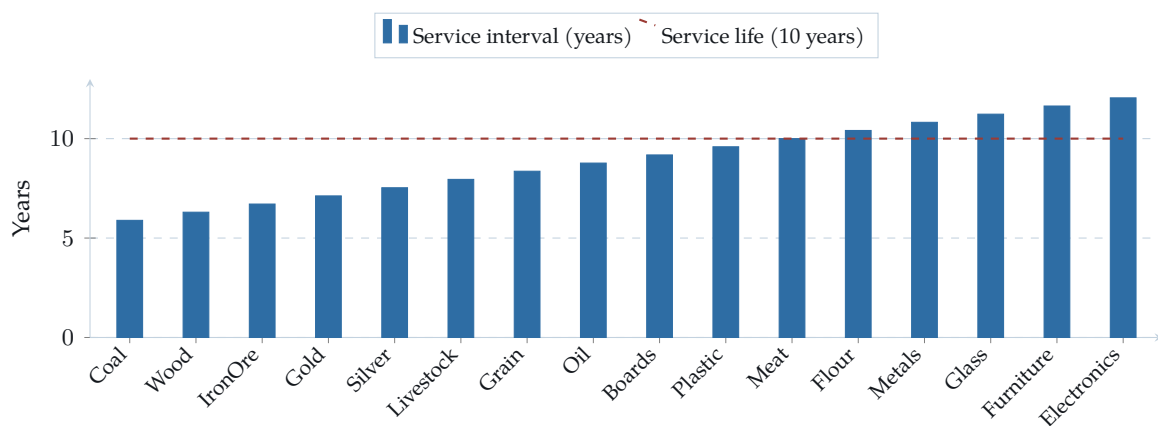


Figure 6 — Road vehicle service interval against the fixed 10-year service life. Six types have an interval longer than their lifetime.

Balance finding: six road vehicles are never serviced. Service life is a uniform 10 game years (3 650 days), but intervals run to 4 400 days. For the Flour, Metals, Glass, Furniture and Electronics trucks — and marginally the Meat Truck at exactly 3 650 — the interval exceeds the lifetime, so the vehicle reaches end of life before its first service. The parameter is dead for over a third of the road catalogue.

Trains are unaffected: Iron Dragon services roughly every 4.1 years against an 8-year life, Desert Runner every 5.5 against 15, Thunderbolt every 6.8 against 20.

7.6 Routes and trading

A vehicle arriving at a station waits 10 seconds. Two seconds after arrival the trade executes, exactly once per stop. The station instance is resolved from the tile through the appropriate registry, with a full rescan as fallback. On success the revenue is added to the balance, recorded in the budget ledger, a cash sound plays and a floating price label appears for 3 seconds.

No trade occurs at the final stop before returning to the depot, during a route interruption, or if no station instance can be resolved for the tile. The trade result carries success, revenue, units loaded, units unloaded and a failure reason.



The depot panel: locomotive and wagon selection, wagon count, route list and the per-vehicle command set.

8 Economy

8.1 Account

STARTING BALANCE	100 000 CR
CURRENCY SUFFIX	CR
STORAGE TYPE	Signed, so the balance may go negative
DISPLAY FORMAT	100.000 CR, thousands separated by a full stop, independent of OS locale

Construction and purchases require cover and are blocked if unaffordable. Recurring monthly deductions — wages and operating costs — are taken regardless and may push the balance below zero.

8.2 Revenue

Cargo delivery is the only source of income; nothing else increases the balance except demolition re-funds. Revenue arises **only from unloading goods into a receiving facility**. Loading raw materials at a producing facility is a transfer and earns nothing.

$$\text{revenue} = \text{round}\left(\frac{\text{base price} \times \text{multiplier} \times \text{units unloaded}}{\text{carrier capacity}}\right)$$

The base price is quoted **per full carrier** — one wagon on rail (capacity 15) or one vehicle on road (capacity 20) — so a partial load is paid pro rata. Because every wagon in a train shares one type and therefore one capacity, the capacity of a single unit is sufficient for the calculation. Rounding is half-up in integer arithmetic, computed as $(\text{price} \times \text{mult} \times \text{units} + \text{capacity}/2) / \text{capacity}$, with

the intermediate held in a 64-bit integer as protection against overflow at high multipliers.

Revenue is zero if the base price, the units unloaded or the capacity is zero or below.

8.2.1 Price list

ID	RESOURCE	PER FULL CARRIER		PER UNIT		ROAD PREM.
		RAIL	ROAD	RAIL	ROAD	
1	Coal	300	500	20.00	25.00	+25 %
2	Wood	300	500	20.00	25.00	+25 %
3	Iron Ore	300	500	20.00	25.00	+25 %
4	Gold	300	500	20.00	25.00	+25 %
5	Silver	300	500	20.00	25.00	+25 %
6	Livestock	350	550	23.33	27.50	+17.9 %
7	Grain	350	550	23.33	27.50	+17.9 %
8	Oil	300	500	20.00	25.00	+25 %
9	Boards	300	500	20.00	25.00	+25 %
10	Plastic	300	500	20.00	25.00	+25 %
11	Meat	350	550	23.33	27.50	+17.9 %
12	Flour	300	500	20.00	25.00	+25 %
13	Metals	400	600	26.67	30.00	+12.5 %
14	Glass	200	300	13.33	15.00	+12.5 %
15	Furniture	400	500	26.67	25.00	−6.3 %
16	Electronics	500	600	33.33	30.00	−10.0 %

Prices fall into bands. On rail: 200 for Glass, 300 for the basic materials, 350 for the foodstuffs, 400 for Metals and Furniture, 500 for Electronics. On road: 300 for Glass, 500 for the basic materials and Furniture, 550 for the foodstuffs, 600 for Metals and Electronics.

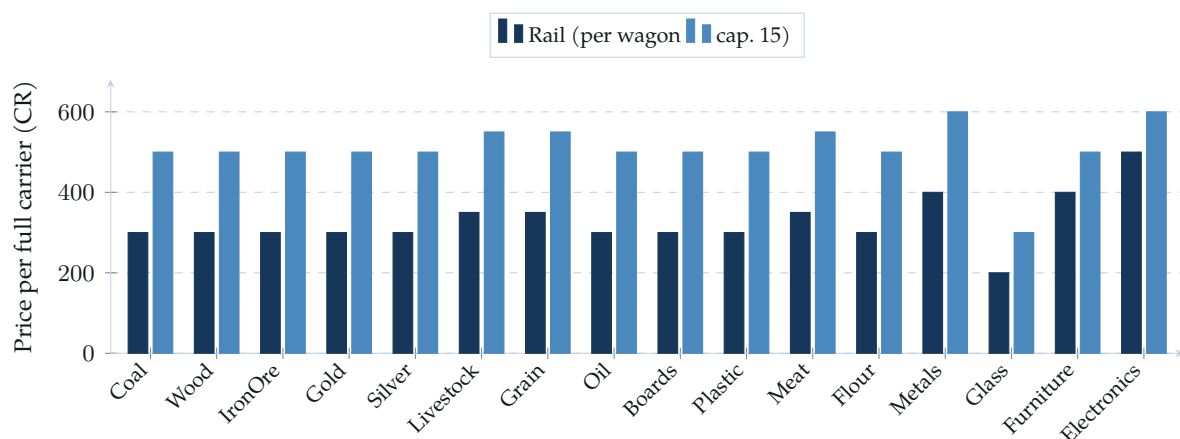


Figure 8 — Base prices per full carrier. Note that the bands are narrow: the most valuable cargo in the game is worth only 2.5 times the least valuable on rail.

8.2.2 Worked examples

DELIVERY	CALCULATION	REVENUE
Rail, Coal, one full wagon (15/15)	$300 \times 15/15$	300 CR
Rail, Coal, half a wagon (7/15)	$(300 \times 7 + 7)/15$	140 CR
Rail, Coal, full 5-wagon train (75)	$300 \times 75/15$	1 500 CR
Rail, Electronics, full 10-wagon train (150)	$500 \times 150/15$	5 000 CR
Road, Coal, one full vehicle (20/20)	$500 \times 20/20$	500 CR
Road, Metals, one full vehicle	$600 \times 20/20$	600 CR
Rail, Coal, 5-wagon train at $\times 3$	$300 \times 3 \times 75/15$	4 500 CR

A misleading comment in the source. ResourcePricing.cs carries an illustrative example using a wagon capacity of 30 and a coal price of 50. Neither matches the live catalogue, where WagonSpec.MaximumCapacity is 15 and the rail price of coal is 300. The table above is authoritative; the comment should be corrected or removed before it misleads someone.

8.3 Dynamic price multiplier

Re-evaluated once per game month, at the month boundary, after settlement.

- **Balance below zero** — escalation. The first negative month sets the multiplier to $\times 2$; each further consecutive negative month adds one ($\times 3$, $\times 4$, $\times 5$, ...).
- **Balance above zero** — reset to $\times 1$.
- **Balance exactly zero** — unchanged.

The multiplier doubles as the counter of the negative streak, so no separate state is needed. It is saved and clamped to a minimum of 1 on load. The design intent is a rubber band: when the player falls into deficit, cargo becomes more valuable so they can climb out.

8.4 Value along the chain

Because revenue is earned on each unloading, a supply chain pays the player once per hop rather than once per finished good. The table below sets the rail price of what enters a processor against the rail price of what leaves it, per full wagon.

PROCESSOR	CONVERSION	IN	OUT	DELTA
Smelter	Iron Ore → Metals	300	400	+100
Sawmill	Wood → Boards	300	300	0
Oil Refinery	Oil → Plastic	300	300	0
Slaughterhouse	Livestock → Meat	350	350	0
Grain Factory	Grain → Flour	350	300	−50
Glass Factory	Metals → Glass	400	200	−200
Power Station	Coal → (consumed)	300	—	—
Electronics Fac.	Plastic, Glass, Metals → Electronics	900	500	−400
Furniture Fac.	Plastic, Glass, Metals, Boards → Furniture	1 200	400	−800

The input figures for the two tier-two factories add one full wagon of each required resource, which overstates them if the conversion ratios differ; those ratios are not yet documented.

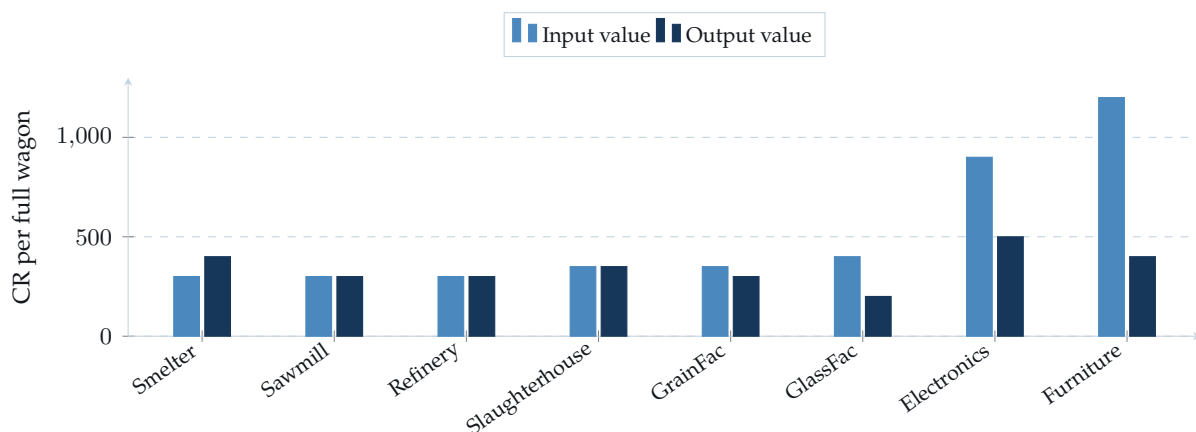


Figure 9 — Input against output value per full wagon. Only the Smelter raises the price of what passes through it.

8.4.1 What this means

Read as a value ladder, the price list looks broken: only one of eight converting processors raises the price of the goods passing through it, and the two tier-two factories appear to destroy value outright.

Read correctly, it is not a ladder at all. **The player is paid for haulage, not for manufacture.** Every delivery into a receiving facility pays, so a long chain is profitable because it contains more paid hops, not because the goods appreciate. Iron Ore into a Smelter, Metals into a Glass Factory and Glass into a Furniture Factory are three separate payments of 300, 400 and 200 — 900 CR in total from one ore deposit, against 300 CR for hauling the ore one hop and stopping.

This is a coherent design, and arguably a better one than a value ladder, because it rewards network depth rather than end-product selection. But it is worth stating deliberately, because two consequences follow. The first is that the price of a good never has to relate to its position in the chain, which frees the numbers but also makes them arbitrary. The second is more awkward:

Six resources appear to have no consumer. Revenue requires unloading into a receiving facility. Cross-referencing the sixteen facilities against the sixteen resources, ten are accepted as inputs somewhere: Coal, Wood, Iron Ore, Oil, Livestock, Grain, Boards, Plastic, Glass and Metals.

Gold, Silver, Meat, Flour, Furniture and Electronics are accepted nowhere. No facility lists any of them as an input, yet they carry some of the highest prices in the game — Electronics is the most valuable cargo at 500 per wagon, and Furniture and Metals sit at 400.

On the documented evidence, the six most obviously desirable end products cannot be sold, and the deepest chains in the game terminate in goods with nowhere to go. Two readings are possible: either the trade systems permit a sale at a station without a receiving facility — in which case the pricing of those six is meaningful and this is a non-issue — or those goods are currently dead ends and the entire tier-two branch of the production graph earns nothing at its final step.

This is the single most important thing to verify in the project. The answer is in `TrainTradeSystem` and `VehicleTradeSystem`, which were not part of the analysed source. Until it is settled, the profitability of every chain beyond the first hop is unknown.

Rail and road are priced in opposition to their capability. Road pays more per unit than rail for fourteen of the sixteen resources, by between 12.5 % and 25 %; only Furniture and Electronics favour rail. Read as a deliberate counterweight this makes sense — rail wins on volume, so road is given a unit premium to stay relevant. The counterweight is however far too small to bite: as §9 shows, a good train moves roughly twelve times what a road vehicle moves, so a premium of a quarter leaves road non-competitive wherever rail can reach.

8.5 Monthly settlement

$\text{outflow} = \text{wages} + \text{train operating costs} + \text{vehicle operating costs}$

Deducted as a single transaction, so the UI sees one balance change. Operating costs come from the locomotive only; wagons carry none.

8.5.1 Order of operations at the month boundary

Determinism depends on this sequence:

1. Settle the elapsed month: wages and full-capacity streak evaluation, then fleet operating costs, then the budget ledger and any annual report, then the price multiplier.
2. Daily factory production, already counted into the new month.
3. Daily fleet servicing and lifetime checks.

The simulation is single-threaded by deliberate choice: the Unity API is not thread-safe, the computational load is negligible, and sequential execution guarantees determinism.

8.6 Annual report

Displayed at the start of each game year, after December has been settled. The first report therefore appears on 1 January 1951 and covers 1950.

- **Income:** transportation revenue, trains and road vehicles combined.
- **Expenditure:** trains purchased with count and total, vehicles purchased, factories purchased, train operations, vehicle operations, employee salaries.

$$\text{Net result} = \text{revenue} - \text{total expenses}$$

Positive results are shown in green with a plus sign, negative in red with a minus, zero neutral. The budget system deducts nothing; it is a ledger only, and resets after display.

8.7 Cost structure in practice

Because every cost is now known, the shape of the early game can be stated exactly. A minimal but complete operation — one extraction site, one processor, about 30 tiles of track, two stations, two depots and a five-wagon Iron Dragon — costs:

ITEM	QTY	COST
Extraction facility	1	500 CR
Processing facility	1	300 CR
Track pieces	30	750 CR
Stations	2	60 CR
Depots	2	100 CR
Iron Dragon + 5 wagons	1	150 CR
Total		1 860 CR
As a share of starting balance		1.86 %

Balance finding: there is no early-game capital constraint, and the first delivery nearly repays the whole setup. With prices known, the payback can be stated exactly. One full five-wagon coal train unloads 75 units for **1 500 CR** — 81 % of the 1 860 CR cost of the entire chain above, including the track, both stations, both depots, both facilities and the train itself. The second delivery puts the operation into profit, roughly two and a half real minutes after it starts running.

The starting balance funds roughly **54 complete supply chains** before the player earns a single credit, or the equivalent of **66 full coal trainloads**. Recurring costs are similarly light: a train costs 5 to 10 CR a month and a road vehicle 3 CR, so a fleet of twenty vehicles costs under 200 CR a month against a balance of 100 000.

Combined with the escalating price multiplier, which makes deficit easier to escape, and the absence of any loss condition, the economy currently applies almost no pressure. Whether that is acceptable depends on the answer to the win-condition question in §14: a pure sandbox may legitimately choose comfort. But if any tension is wanted in the first hour, the starting balance is by far the cheapest lever — reducing it to the low thousands would make the first chain a real decision without touching another value.

9 Throughput analysis

The figures below are derived from the documented constants and are the main tool for reasoning about how many vehicles a network needs. They assume level ground, ignore acceleration — which, per §7.4, is constant — and count the two 10-second station waits per round trip.

$$\text{cycle time} = \frac{2D}{v} + 20 \text{ s}, \quad \text{throughput} = \frac{\text{capacity}}{\text{cycle time}}$$

where D is the one-way distance in tiles and v the speed in tiles per second.

	CAP.	D	CYCLE (S)	UNITS/MIN	CR/MIN
CONFIGURATION					
Producer at level 2 (reference)	—	—	—	200.0	—
Producer at level 1	—	—	—	120.0	—
Producer at level 0	—	—	—	60.0	—
Thunderbolt, 10 wagons	150	20	73.3	122.8	2 456
Thunderbolt, 10 wagons	150	50	153.3	58.7	1 174
Iron Dragon, 5 wagons	75	20	153.3	29.3	586
Iron Dragon, 5 wagons	75	50	353.3	12.7	254
Road vehicle, speed 80	20	20	120.0	10.0	250
Road vehicle, speed 45	20	20	197.8	6.1	153

Revenue assumes Coal, at 20 CR per unit on rail and 25 CR on road. Hauling Electronics instead, at 33.33 CR per unit on rail, raises the best train configuration to roughly 4 090 CR per minute.

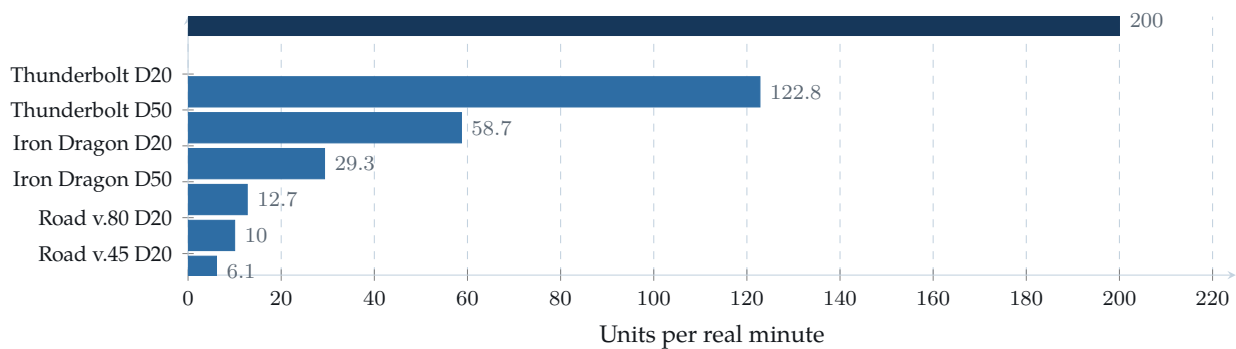


Figure 7 — A single level-2 producer outproduces every available single-vehicle configuration. The best train in the game, on a short line, moves 61 % of what one mine generates.

9.1 What this implies

- **Producers outrun transport.** A level-2 extraction site needs between two and seven vehicles to be fully served, depending on distance and equipment. Serving a site properly is therefore a real logistical problem, which is a good outcome for the design.
- **The self-limiting effect is benign.** A producer that fills up simply stops, and because no slot changes, no wage is charged. An under-served mine costs nothing to own, so the penalty for under-capacity is opportunity cost rather than bleeding.
- **Storage caps rarely bind.** At level 2 a 10 000-unit slot takes 1 000 game days to fill — roughly 2.7 game years, or 50 real minutes of complete neglect. The caps are a long-session backstop rather than an active constraint.
- **Road transport is a short-haul tool.** At 20 units per trip, road vehicles run at roughly one twelfth of a good train. Their place is feeder work and short links rather than trunk haulage, which is a coherent role and worth making explicit to the player.
- **Distance punishes slow stock.** Raising the haul from 20 to 50 tiles cuts an Iron Dragon's throughput by 57 %. The Thunderbolt's speed advantage matters most precisely on the long lines where the Iron Dragon collapses.
- **Operating costs are rounding error.** A Thunderbolt costs 10 CR a month, or about 6.7 CR per real minute, against roughly 2 456 CR per minute of gross revenue on a short coal line — under 0.3 %. Once a line is running, the recurring cost side of the economy stops being a consideration at all.

10 User interface

10.1 HUD

A persistent bottom bar carries eight buttons: rail construction, road construction, stations, trains, road vehicles, map, factories and staff. To their right sit the current month and year, the balance in CR, zoom controls and the game menu.

10.2 Panels

RAIL CONSTRUCTION	Terrain tools, rail pieces, depots, stations
DEPOT RAIL	Train configuration, selected-train status, route management, vehicle commands
FACTORIES	Producer and processor catalogue
STAFF MANAGEMENT	Hire Staff and Staff Management modes, profession list, portrait, salary tiers
MAP	Terrain, Industry, Rail and Road views with legends
ANNUAL REPORT	Yearly income and expenditure statement with coloured net result

Panels are non-modal and several can be open at once.

10.3 Map overlays

Terrain View legend: five elevation bands, water, towns, railways, rail stations, roads, road stations.
Industry View legend: a distinct colour per facility type, grouped into extraction and processing.
Rail View and **Road View** show each network alone. A white rectangle on the minimap marks the camera position.

10.4 Feedback

City and station names are drawn as world-space labels at height 10.0 with font size 6.0. A successful trade plays a cash sound and shows a floating price label for 3 seconds. Facility construction shows a percentage label that lingers 2 seconds after completion. The annual report plays its own sound.



The map screen with Terrain View selected, showing the view switcher and both legend blocks.

11 Art direction and presentation

The game combines a licensed low-poly 3D art base (Synty Studios / Unity Asset Store) with a custom-built pixel-art interface layer. This is a deliberate production decision for a solo project: engineering effort goes into systems depth while a coherent commercial art library carries the world.

The camera is a fixed-angle isometric view over a diamond-tile grid with zoom control; the build cursor is a white tile outline, so the grid is explicit during construction and invisible during play. Buildings are simple stacked volumes; city buildings may be authored prefabs or generated primitives with five colour variants per tier. Terrain vertices are coloured by height: black below 1.5, red from 1.5 to 3.5, green above 3.5.

Infrastructure always reads darker than terrain, so the network is the first thing the eye follows. Rail and road are both offset 0.3 above the terrain surface. The tone is bright, daylit and calm, with no weather and no day/night cycle.

12 Audio

Three cues are confirmed in the systems examined: a cash sound on a successful trade, an annual report sound, and error feedback through the status error menu.

Otherwise the audio design is undocumented: music tracks and their sourcing [TBD]; ambient loops [TBD]; vehicle and industry sound effects [TBD]; mixing and volume controls [TBD]; and whether assets are licensed or original [TBD].

13 Technical specification

ENGINE	Unity
TARGET PLATFORM	Windows 10/11, 64-bit
GRAPHICS API	DirectX 11/12
DISTRIBUTION	Steam (App ID 5155330)
SOURCE FILES	86 total, of which roughly 20 are main source files
THREADING	Single-threaded by design; no Job System
PATHFINDING	A* with a 10 000-iteration cap and 0.4 s debounce
PERSISTENCE	Save API present (IndicatrixAPI); the price multiplier is persisted and clamped on load
SETTINGS	[TBD: document resolution, graphics and audio options]
LOCALISATION	English only; are strings externalised? [TBD]
ACHIEVEMENTS	[TBD: confirm Steam integration]

13.1 Development history

JUL 2024	Project begins as a part-time hobby project
17 MAR 2025	Initial architecture created
TO 17 JAN 2026	Roughly 40 % of development completed without AI assistance: core architecture, tile system, terrain rendering
FROM JAN 2026	AI adopted as a programming assistant (ChatGPT, Claude Sonnet / Opus), roughly 50 prompts in total
APR--JUN 2026	Most intensive development period
AUG 2026	Game completed

AI-assisted code is present in the codebase and is disclosed on the Steam store page in line with Valve’s requirements. All design, systems, procedural generation and interface decisions are the developer’s own; all world art consists of licensed commercial assets.

14 Open design questions

14.1 Whether six resources can be sold at all

Status: needs verification, critical. See the warning in §8.4. Gold, Silver, Meat, Flour, Furniture and Electronics are not accepted as an input by any facility, yet revenue is documented as arising only from unloading into a receiving facility. Either the trade systems allow a sale without one, or the game’s six most valuable end products are dead ends. Nothing else in this document matters as much for the economy.

14.2 Objectives and win conditions

Status: open. The game has no defined objective or win condition. A sandbox with no end state is legitimate — OpenTTD has sustained an audience for two decades on that basis — but it makes it harder for a new player to know whether they are doing well, and harder for a reviewer to describe what the game asks of them.

The annual report is a strong existing hook. It already computes a net result per year; a cumulative company rating, a target, or a scored end-of-century summary could be layered on it without removing the sandbox.

14.3 The salary level is a dominant strategy

Status: open, newly identified. See §6.4. Level 2 returns 67 % more production per credit of wage than level 0, so no reason exists to choose otherwise.

14.4 Whether production is free

Status: needs verification, potentially critical. See the second warning in §6.4. If the interface permits `EmployeeSalary = 0` alongside `LevelSalary = 2`, maximum production costs nothing and the wage system can be bypassed entirely.

14.5 The acceleration model is inert

Status: open, newly identified. See §7.4. No configuration can exceed the acceleration floor, so locomotive power and wagon weight have no effect. Either the tuning constant should rise or the floor should fall — or, if constant acceleration is intended, the weight and power fields should be removed rather than implying a system that does not exist.

14.6 Six road vehicles are never serviced

Status: open, newly identified. See §7.6. Service intervals above 3 650 days exceed the 10-year vehicle lifetime.

14.7 Economic pressure

Status: open. The starting balance funds roughly 54 complete supply chains, one coal trainload repays 81 % of a first chain, operating costs run below 0.3 % of gross revenue, and the price multiplier assists recovery from deficit. There is very little economic pressure at any point, and effectively none after the first delivery. See §8.7 and §9.

14.8 The role of cities

Status: open or undocumented. Nothing in the analysed systems shows cities consuming, demanding or growing. If a city is only a delivery location and a source of station names, that should be a stated decision.

14.9 Whether time carries mechanical weight

Status: open. Locomotives and vehicles carry a year of manufacture — 1930 to 2000 for trains, 1967 to 1997 for road vehicles — which strongly suggests date-gated availability was intended. Whether it is implemented is not established. If it is not, the 1950 start date and the entire year field are decoration; if it is, it is an unmentioned selling point.

14.10 Whether the price list should track the chain

Status: open, newly identified. Because the player is paid per hop rather than per value added, prices need not rise along the chain — and mostly they do not. Only the Smelter raises the value of what passes through it; the Glass Factory halves it. This is defensible, but it means a player who reasons about the price list as a progression will be misled. Either the prices should be brought into line with chain depth, or the design should make clear that depth pays through the number of deliveries rather than through the value of the goods.

14.11 Glass is made from Metals

Status: open, minor. An unusual chain that players will notice, and one that halves the price per wagon (400 to 200). Either accept it as a stylised abstraction or revisit it.

14.12 Passenger transport

Status: closed for this version. Cargo only, by design. Listed because it is a standing genre expectation.

Appendix A • Quick reference

Starting balance	100 000 CR
Game start	January 1950
Time base	3 real seconds = 1 game day
Grid	256 × 256 tiles
Cities per map	25–30
Track, curve, switch, crossroad	25 CR
Station	30 CR
Depot	50 CR
Terrain level up / down	10 CR
Extraction / processing facility	500 CR / 300 CR
Demolition refund	50 % of base cost, rounded down
Locomotives	100 / 150 / 200 CR
Wagon	10 CR, capacity 15, max 10 per train
Road vehicle	50 CR, capacity 20
Train operating cost	5 / 7 / 10 CR per month
Vehicle operating cost	3 CR per month
Production by level	+3 / +6 / +10 units per day
Decay	–10 per day after 3 full months
Wage multiplier by level	×1.0 / ×1.5 / ×2.0
Station wait / trade delay	10 s / 2 s
Price multiplier while negative	×2, ×3, ×4 ...escalating monthly
Station / factory exclusion radius	2 / 3 (Chebyshev)
Sale formula	round(price × multiplier × units unloaded / capacity)
Rail price per wagon (cap. 15)	200 Glass • 300 basic • 350 food • 400 Metals, Furniture • 500 Electronics
Road price per vehicle (cap. 20)	300 Glass • 500 basic, Furniture • 550 food • 600 Metals, Electronics
Full 5-wagon coal delivery	1 500 CR
Full 10-wagon electronics delivery	5 000 CR

Appendix B • Remaining gaps

The value analysis closed most of what version 0.1 was missing. What remains falls into three groups.

Priority 1 — blocks economic balancing

- **Trade rules** (`TrainTradeSystem`, `VehicleTradeSystem`) — which station slots are served and in what order, how many units transfer at once, the success conditions, and above all **whether a sale is possible without a receiving facility**. This last point decides whether six of the sixteen resources have any value at all. See §8.4.
- **Conversion ratios** — how many input units a processor consumes per output unit produced. Without these the tier-two factories cannot be evaluated, since their input value can only be bounded, not calculated.
- **Station catchment radius** (`StationInstance` and the two registries) — how many tiles around a station are searched for facilities. This determines how tightly a network can be packed.

Priority 2 — needs a decision rather than a lookup

- Whether the interface permits a zero employee salary at level 2 (§6.4)
- Whether the acceleration model should be made to function (§7.4)
- Whether road vehicle service intervals should be brought under the 10-year lifetime (§7.6)
- Whether the starting balance should create early pressure (§8.7)
- Whether year of manufacture gates vehicle availability (§14.8)

Priority 3 — completes the record

- Environment generation: tree, rock and landing location density (`EnvironmentManager`)
- Contents of the statistics screens
- Audio design in full (§12)
- Settings, localisation readiness, achievements (§13)
- Camera controls
- Reconciliation of sea floor level -2 against minimum elevation -1 (§3.1)

A standing caveat. Every number in this document is the default found in code. Fields marked `[SerializeField]` can be overridden in the Unity Inspector, and Unity keeps previously serialised values when a script is replaced — a hazard the project's own source comments flag explicitly. Before any of these values is used for balancing work, it should be confirmed against the live scene.

Appendix C • Interface reference

Screens captured from the current playable build at 1920 × 1080.



Rail Construction and Depot Rail open simultaneously over a dense junction. Panels are non-modal and can be combined.



Rail construction with the tile build cursor on the grid.



Road and rail networks serving the same region.



Two loaded coal services crossing on intersecting lines.



City scale variation and a road corridor operating independently of rail.



Generated terrain showing the object types that constrain facility placement.